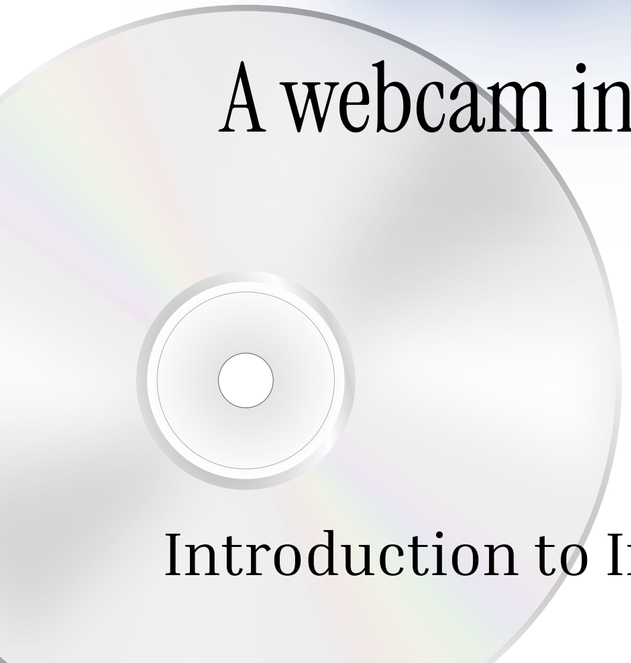
Alyssa, Guy, Hider, Armin

HANDMONICA

A webcam instrument for melody, harmony, and guided practice



Introduction to Image Recognition AI and Robotics Lab | Final Project Checkpoint

Musical Exploration with a Webcam

ULTIMATE GOAL

Make playing and learning music as easy as raising your hand: a laptop and a webcam become an instrument and a practice tutor

Beginners & Kids

Play a real melody within minutes

Without an Instrument

Only a webcam is needed

Music Learners

Practice reading numbered notation

Why it matters?

It lowers the first barrier to music: no money for an instrument, no finger techniques, and it teaches numbered notation while you play

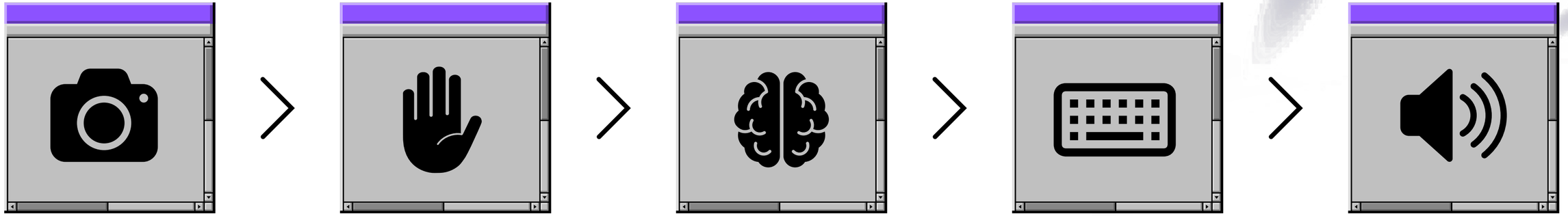
The first steps of music are hard

For Learners

- Instruments cost money and are not always available
- Finger technique takes weeks before anything sounds like music
- Reading notation is a separate skill learned in isolation
- Practicing alone gives no feedback on what was wrong

For the technology

- Signs must be recognized in real time, or the music feels delayed
- Every person forms the same sign differently, especially the thumb
- Even with 3 people's data, sign 7 is found only 78% of the time (read as 6)
- Simple finger-counting rules break when the hand tilts or turns



Webcam
Frame

MediaPipe
21 landmarks / hand

Classifier (SVM)
sign 1-7 / rest

Music Logic
note, octave, chord

Piano synth
real-time sound

Three Pillars : One Real-Time Pipeline

1 ***Recognition Model***
The Foundation

2 ***The Instrument***
The Impact Driver

3 ***User Experience***
Screen Guidance & Sharing

Recognition Model : The Foundation

What it does

- MediaPipe finds 21 hand landmarks
- Normalized: wrist at origin, scaled by hand size
- Classifier: 8 classes (rest and 1 to 7)
- Smoothing over frames stops flicker

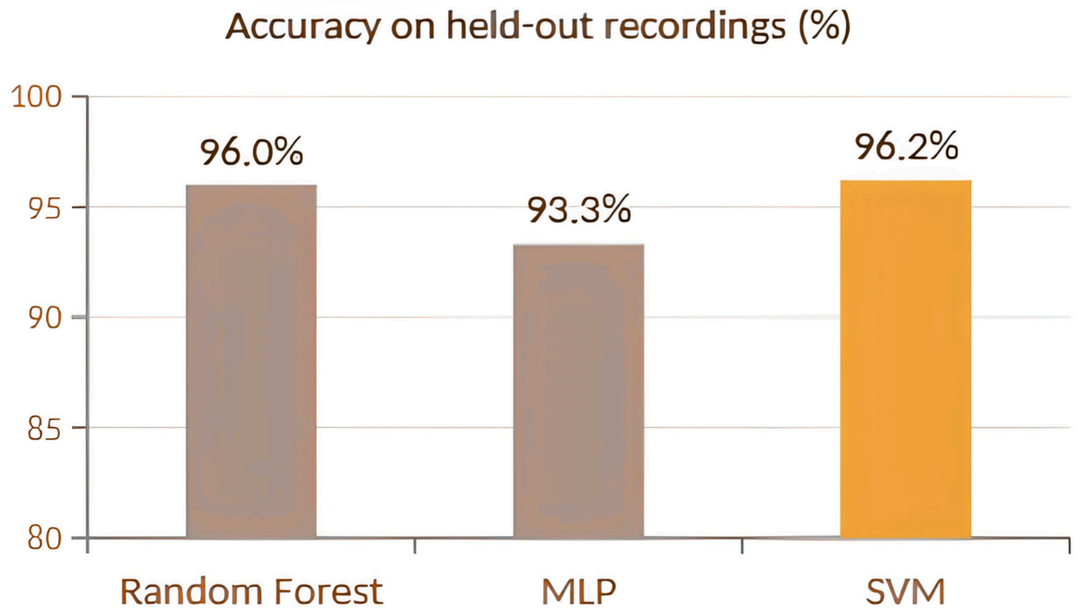
Progress

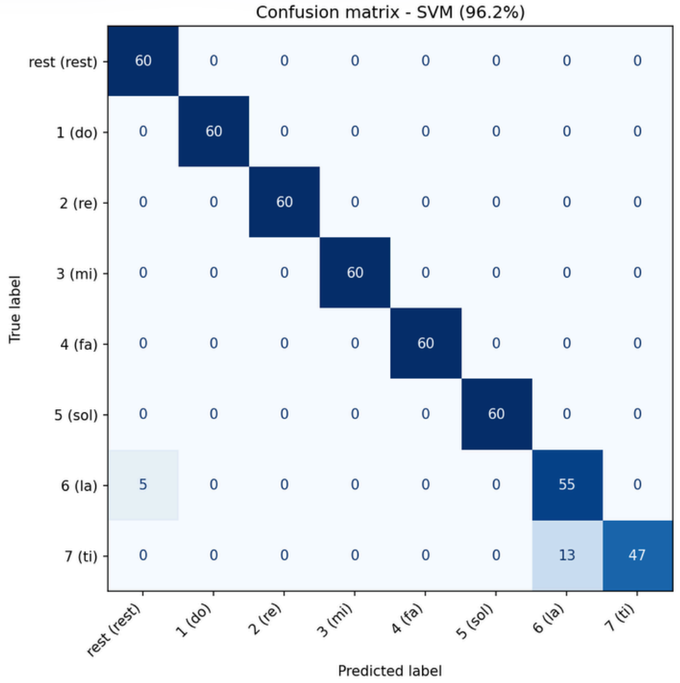
- Own dataset: 6,300 samples from 3 people
- Compared Random Forest, MLP and SVM
- SVM chosen: 96.2% on held-out recordings
- Added sign variations to fix thumb errors

Next step

- Leave-one-person-out test (new users)
- Fix remaining 7 vs 6 confusion
- Compare with rule-based finger counting
- Compare with an image CNN on hand crops

Training Results: SVM reaches 96.2%





96.2%
known users

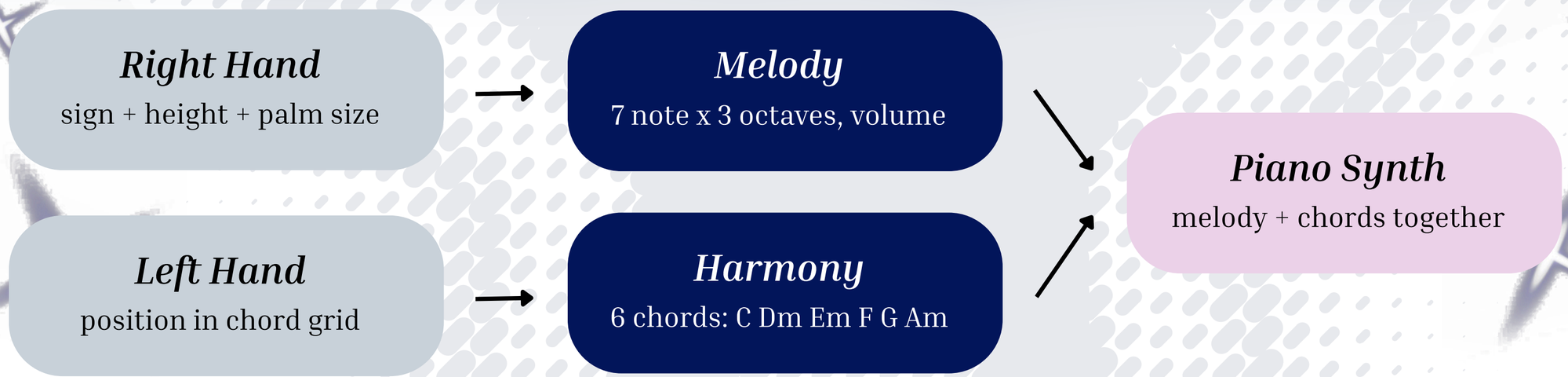
next
new users

6,300 samples - 3 people - 8 classes

Test: one held-out recording burst per sign (480 samples). All 3 people are in training, so this measures known users.

6 of 8 classes perfect

Left: 7 read as 6 (13 of 60), 6 read as rest (5). Still the thumb

The instrument: the impact driver

Progress

- Working real-time prototype: plays a full song
- Flick down to repeat the same note
- Transpose to any key

Next Step

Measure latency: time from sign to sound

User Experience : Guidance

Chord Boxes

Octave zones

Confidence & Volume

Install from GitHub

What it does

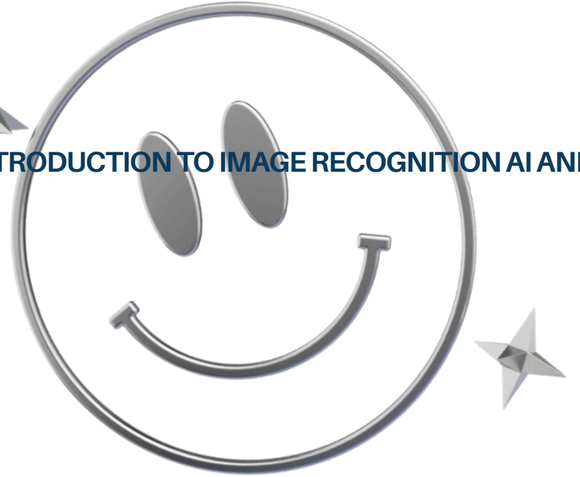
- Shows the player what the model sees
- Chord boxes, octave zones, current note
- Confidence and volume bars
- Anyone can install it from GitHub

Progress

- On-screen display working
- Code + step-by-step README on GitHub
- Tested by 3 people

Next step

- Learning mode : learn each sign step by step, with pictures
- Song mode: show the notation and chords on screen to play along
- User test with beginners by Oct 29



Live Demo

Mary Had a Little Lamb

Right Hand : Melody

Left Hand : Chords

Expression

move closer for louder, higher for a higher octave

NOW

Working real-time instrument with melody + chords

NEXT

More people's data, leave-one-person-out, rule-based & CNN comparison

THEN

Learning mode + song mode, latency, beginner user test (by Oct 29)

FINAL

Final presentation: evaluated model and user-test results

On track: next, prove that it works

Goal for the final: not just an instrument that works, but evidence that it works for new people and helps beginners learn.

Thank You!